Adding and subtracting polynomials

1 Consider the term $9x^8$.

- **a** Identify the coefficient of $9x^8$.
- **b** State the exponent.

2 Consider the expression $5y^3$.

- **a** Determine the number of terms in the expression.
- **b** Identify the coefficient of the term.

3 Consider the expression $-2a^3$.

- **a** State whether this expression is a monomial.
- **b** State the variable, the coefficient, and the degree.

4 For each of the following expressions:

- **i** Determine the number of terms in the expression.
- **ii** Identify the coefficients of the terms.

- **a** $-5y^9 - y$
- **b** $x + 7x^6 - x^9$

5 State whether each of the following is a monomial:

- **a** $2x^3 - 4x^5 + 3$
- **b** $3x^3 + \dfrac{2}{x^7} - 1$
- **c** $\sqrt{2}x^6 + 7x$

6 State whether each of the following statements is true or false.

- **a** The degree of a polynomial in x is the largest coefficient of any of the terms of the polynomial.
- **b** The degree of a polynomial is not always the exponent of the first term.

7 For each of the following polynomials, iden

i The degree

ii The leading coefficient

iii The constant term

a $P\left(x\right) = 2x^7 + 2x^5 + 2x + 2$

b $P\left(x\right) = 6 - \dfrac{7}{10}x^3$

c $P\left(x\right) = \dfrac{x^7}{5} + \dfrac{x^6}{6} + 5$

8 Simplify the following:

a $\left(-4x^2 + 7x - 9\right) + \left(6x^2 + 7x + 8\right)$

b $\left(-8x^2 - 3\right) + \left(-4x^2 - 8x\right)$

c $\left(3x^3 - 9x^2 - 8x - 7\right) + \left(-7x^3 - 9x\right)$

d $\left(\dfrac{1}{2}x^2 - \dfrac{1}{4}x + \dfrac{1}{2}\right) + \left(\dfrac{1}{3}x^2 + \dfrac{2}{5}x - \dfrac{1}{4}\right)$

e $\left(-0.2 - 9.5x - 1.4x^2\right) + \left(2.6 - 7.8x + 7.7x^2\right)$

f $\left(4x^2y - 9xy^2 + 4\right) + \left(-7x^2y + 4xy^2 + 8\right)$

g $\left(3x^2y^2 - 2xy^2 + 4y^2\right) + \left(-9x^2y^2 + 3xy^2 - 8y^2\right)$

h $\left(3x^3 - 3x^2 - 8\right) + \left(-5x^3 + 9x^2 + 1\right)$

i $3\left(-2x + 7\right) - \left(7x^2 + 8x - 5\right) + 7x$

j $-6x^3 + 4x^2 + 4x - \left(3x^3 + x + 1\right)$

k $\left(-5x^2 + 6x - 4\right) - \left(-7x^2 + 7x + 9\right)$

l $\left(-2x^3 + 7x^2 + 9x - 2\right) - \left(9x^3 + 7x^2\right)$

m $\left(4x^2y - 5xy^2 - 4\right) - \left(-8x^2y - 6xy^2 - 9\right)$

n $\left(-x^3 - 3x^2 - 4\right) - \left(x^3 - 9x - 9\right)$

o $\left(\dfrac{2}{3}x^2 + \dfrac{1}{2}x - \dfrac{2}{3}\right) - \left(\dfrac{1}{4}x^2 - \dfrac{2}{3}x + \dfrac{1}{4}\right)$

p $\left(7.6x^2 - 0.5x + 7.3\right) - \left(1.4x^2 - 3.4x - 9.6\right)$

q $\left(x^3 - 4x^2 + 4\right) + \left(8x^2 - 2\right) - \left(-2x^3 + 4x^2 + 8x + 9\right)$

r $\left(-x^2 + 8x - 6\right) + \left(4x^2 - x + 7\right)$

9 If a picture frame has a length of $8x^2 - 9x + 3$ and a width of $6x^3 + 9x^2$, form an expression for the perimeter of the rectangular picture frame.

10 Nadia's hat manufacturing business sells its [redacted] exclusively through two retailers. The profit generated through selling at Just Stuff is modelled by $A(q) = -0.5q^2 + 22.5q + 510$ and the profit generated through selling at Glorious Gifts is modelled by $B(q) = 2.9q^2 + 36.5q + 70$, where q is the number of hats sold.
Form an expression for the polynomial, $C(q)$, that models Nadia's total profit.

11 The revenue generated by Tricia's seafood restaurant is modeled by $R(m) = -3.2m^2 + 10.9m + 990$ and the profit of her restaurant is modeled by $P(m) = 2.3m^2 - 29.3m + 830$, where m is the number of meals produced.
Find the polynomial that models the costs of Tricia's restaurant.

12 If $P(x) = -x^2 - 3x - 6$ and $Q(x) = 6x - 7$, form a simplified expression for $P(x) + Q(x)$.

13 If $P(x) = -5x^2 - 6x - 6$ and $Q(x) = -7x + 7$, form a simplified expression for $P(x) - Q(x)$.

14 If $A(x) = -5x^2 + 5$, $B(x) = -7x + 6$ and $C(x) = -2x^2 - 2x$, form a simplified expression for $A(x) + B(x) + C(x)$.

15 If $A(x) = 6x^2 - 6$, $B(x) = 3x - 3$ and $C(x) = -2x^2 + 7x$, form a simplified expression for $A(x) - B(x) - C(x)$.

16 If $A(x) = -6x^2 - 6$, $B(x) = 4x - 7$ and $C(x) = x^2 - 5x$, form a simplified expression for $B(x) - A(x)$.

17 Consider the polynomials $P(x) = 7x^2 + 4x - 8$ and $Q(x) = 5x - 7$.

a Determine the degree that $P(x) - Q(x)$ would have.

b State what type of polynomial would $P(x) - Q(x)$ be.

c Solve for $P(x) - Q(x)$.

18 $R(x)$ is the result of adding polynomials $P(x)$ and $Q(x)$, where $P(x) = 3x - 6x^2 + 7$ and $Q(x) = 2x^3 - 4x - 3$.

a Express $R(x)$ in simplest form.

b Determine the degree of $R(x)$.

c Identify the leading coefficient of $R(x)$.

d State whether $R(x)$ is a polynomial.

19 Consider the the sum of polynomials $P(x)$ [illegible] -5 and $Q(x) = 2x^6 + 4x^4 + 3$. Identify:

a The degree of the sum of the polynomial.

b The constant term of the sum of the polynomial.

20 Consider the polynomials $P(x) = 4x^3 + 4$ and $Q(x) = 8 - 4x^3$.

a Write the simplified expression for $P(x) + Q(x)$.

b What is the degree of $P(x) + Q(x)$?

c State whether $P(x) + Q(x)$ is classified as a polynomial.

21 If $P(x) = 3x^3 - 7x^2 - x + 2$, $Q(x) = 3 - 9x^3$ and $R(x) = -3x^2 + 9x - x^3 + 6$, find an expression for each of the following in simplest form:

a $P(x) - R(x)$

b $P(x) - (R(x) - Q(x))$

22 State the degree of the resulting polynomial in each case.

a $P(x) + Q(x)$, where $P(x)$ has degree 4 and $Q(x)$ has degree 6.

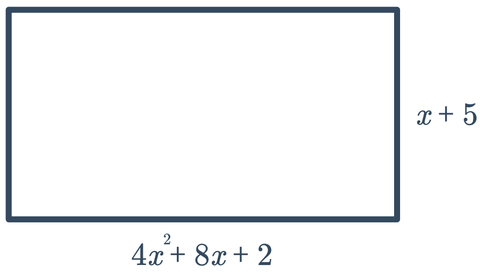
b $P(x) - Q(x)$, where $P(x)$ has degree 4 and $Q(x)$ has degree 6.

c $P(x) - Q(x)$, where $P(x)$ has degree 6 and $Q(x)$ has degree 6, and the leading coefficient of $P(x)$ is not equal to the leading coefficient of $Q(x)$.

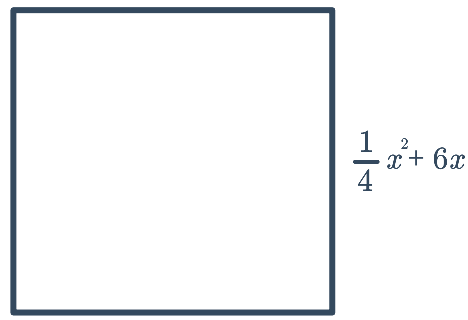
23 Let $P(x) = ax^2 + (b-5)x - 1$ and $Q(x) = x^2 - 5x + 2$.
If $R(x) = 2x^2 + 2x + 1$ is the result of adding $P(x)$ and $Q(x)$, find the values of a and b.

24 Find a polynomial that represents the perim[redacted]f each of the following polygons.

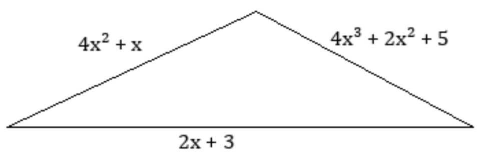
a



b



c



Additional questions

26 Consider the expression $5 + 5x^2 + 13 + \frac{x^2}{5} + \frac{1}{5} + 2x$.

Determine whether each of the following terms could be combined with 5.

a $2x$

b $\frac{1}{5}$

c 13

d $\frac{x^2}{5}$

27 For each of the following expressions:

i Identify a pair of like terms.

ii Simplify the expression by combining these like terms.

a $4y + 9 + 8x + 5$

b $6x + 4 - 5x^2 + 7x$

c $4a + 2b - 6a - 3ab$

d $4xy - 4y^3 + 9x^3 - 4 - 7y^3 + 9x^3y$

28 Determine whether the following statements represent a correct definition for a polynomial.

- **a** A polynomial is the sum of terms of the form ax^n, where a is any real number and n is any real number greater than or equal to zero.
- **b** A polynomial is the sum of terms of the form ax^n, where a is any whole number and n is any whole number.
- **c** A polynomial is the sum of a finite number of terms of the form ax^n, where a is any real number and n is a whole number greater than or equal to zero.
- **d** A polynomial is the sum of a finite number of terms of the form ax^n, where a is a whole number and n is any real number.

29 Create a binomial expression with a degree of 4 and a leading coefficient of 2.

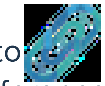
30 Create a monomial expression with a degree of 0.

31 Fill in the blank with a single term that satisfies the given condition
$4x^3 - 5x + \square$

- **a** Creates a trinomial with a leading coefficient of 5
- **b** Creates a binomial with a leading coefficient of 2
- **c** Creates a trinomial with a degree of 3
- **d** Creates a polynomial with a degree of 1

32 Without repeating any values, fill in the blanks using only single-digit numbers to create a polynomial
$\square x^{\square} + \square x^{\square} - \square x^{\square} + \square x + \square$

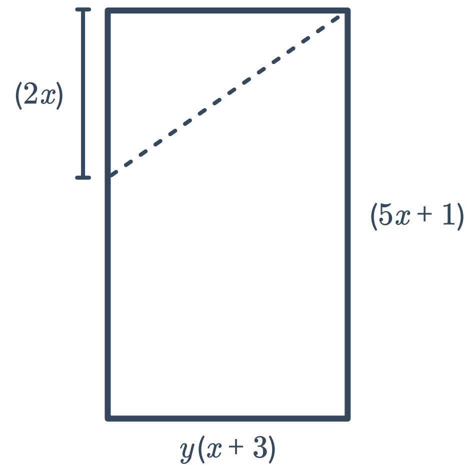
- **a** With the lowest possible degree.
- **b** With the highest possible leading coefficient.
- **c** With the minimum number of terms.

33 A rectangle with the given dimensions is to  a right triangle cut out from one corner. Write and simplify a polynomial sum or difference to model each of the following:

a An expression for the length represented by y.

b An expression for the perimeter of the rectangle before the triangle has been removed.

c If the area of the rectangle is $5x^2 + 16x + 3$ and the area of the triangle is $x^2 + 3x$, determine the area leftover once the triangle has been removed.



34 Explain how two trinomials can be added together to produce a binomial.

35 A polynomial of degree m and a polynomial of degree n, with $m \geq n$, are added together. What is the highest possible degree of the result?

36 Fill in the blanks to make each equation true.

a $(x^3 - x + 7) + (\square x^3 + \square x + 3 = 10$

b $(\square x^2 - 5x - 5) - (\square x + 3) = 4x^2 - 2x - 2$

c $(\square x^{\square}) + (2x^3 - 3x + 1) = 2x^3 + x + 1$

d $(4x + 1) - (\square x^{\square} + \square) = 0$

37 A piece of paper with dimensions 8.5 inches by 11 inches will have a square with length x cut out from each corner to form a box.

Write and simplify a polynomial sum that can be used to represent the perimeter of the shape formed once the corners have been removed.

38 Determine whether each statement is always, sometimes, or never true. Explain your reasoning with examples.

a Two polynomials added together will result in a polynomial.

b A linear function is a polynomial.

c A polynomial added to a non-polynomial will result in a polynomial.

d A non-polynomial added to a non-polynomial will result in a polynomial.